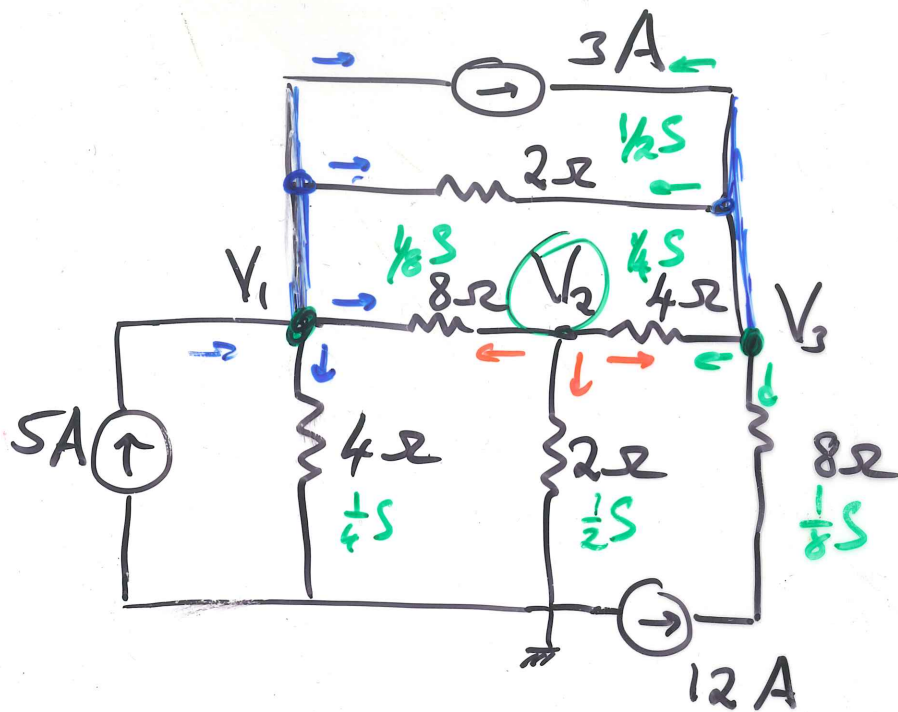




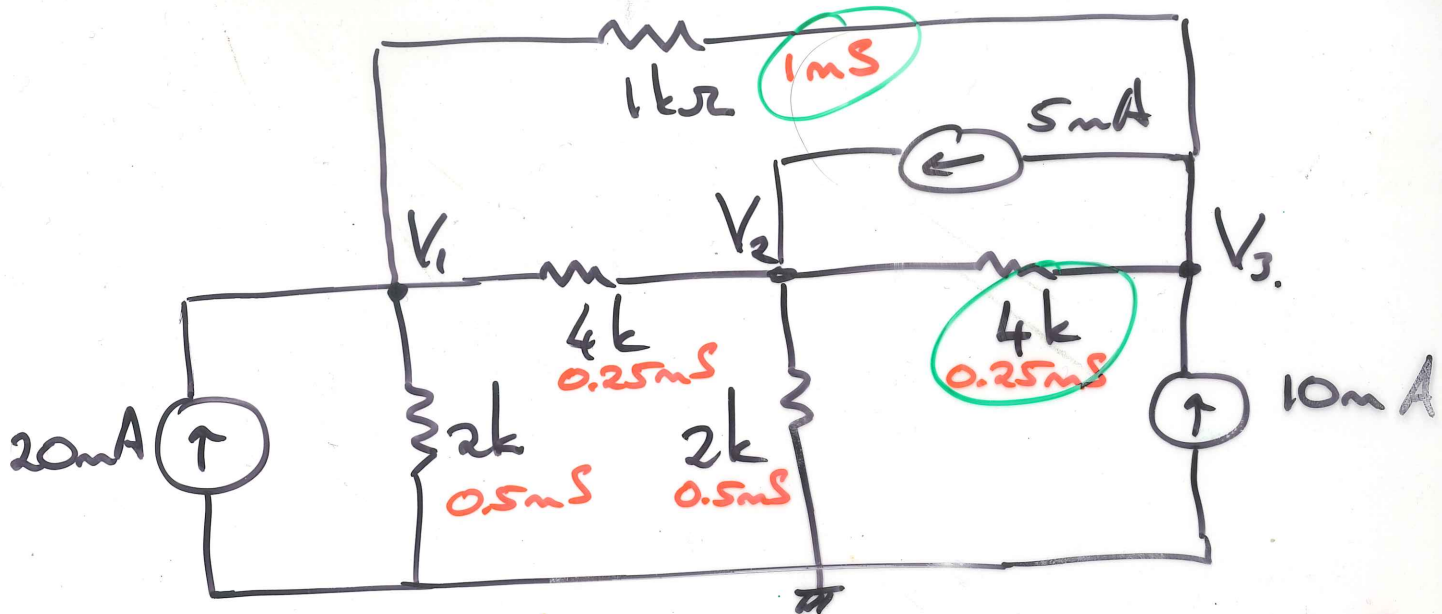
$$\textcircled{1} \quad 5 = 3 + \frac{V_1 - V_3}{2} + \frac{V_1 - V_2}{8} + \frac{V_1 - 0}{4}$$

$$\textcircled{2} \quad 0 = \frac{V_2 - 0}{2} + \frac{V_2 - V_3}{4} + \frac{V_2 - V_1}{8}$$

$$\textcircled{3} \quad \sum I_{IN} = \sum I_{OUT}$$

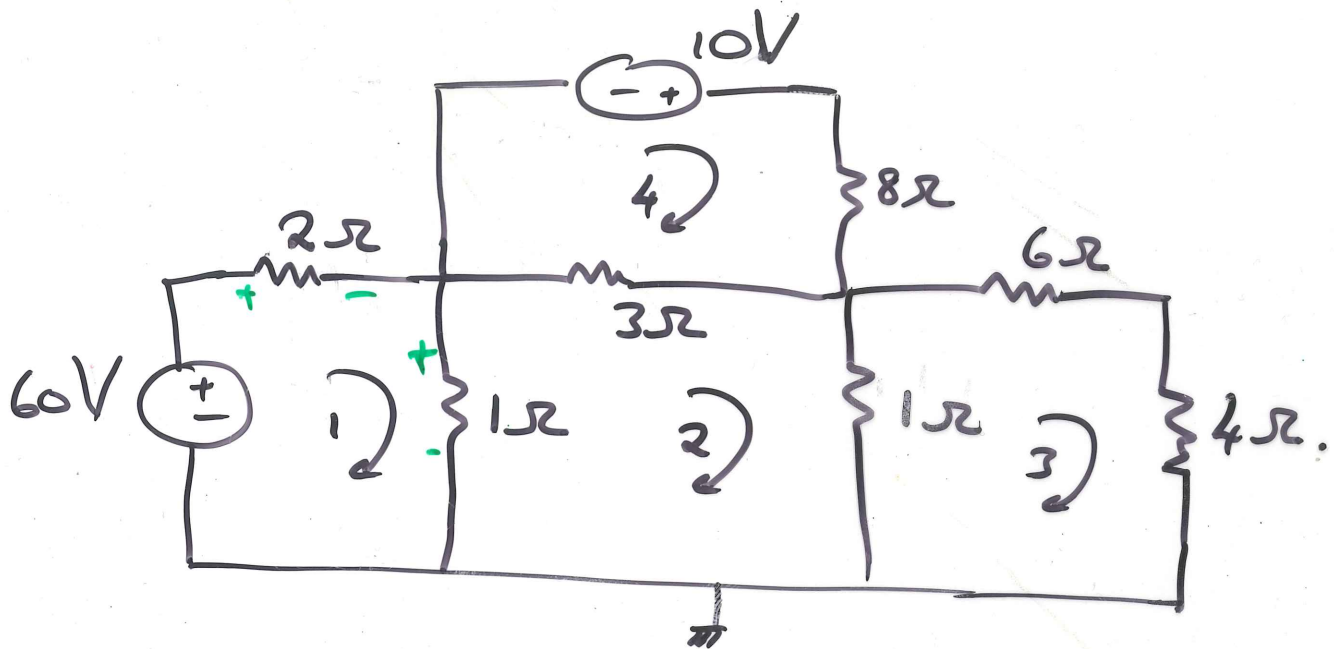
$$12 + 3 = \frac{V_3 - V_1}{2} + \frac{V_3 - V_2}{4}$$

$$\begin{aligned} \rightarrow 5 - 3 &= V_1 \left(\frac{1}{2} + \frac{1}{8} + \frac{1}{4} \right) - V_2 \left(\frac{1}{8} \right) - V_3 \left(\frac{1}{2} \right) \\ 0 &= -V_1 \left(\frac{1}{8} \right) + V_2 \left(\frac{1}{2} + \frac{1}{4} \right) - V_3 \left(\frac{1}{4} \right) \\ 12 + 3 &= -V_1 \left(\frac{1}{2} \right) - V_2 \left(\frac{1}{4} \right) + V_3 \left(\frac{1}{2} + \frac{1}{4} \right) \end{aligned}$$



$$G_s = \frac{1}{2k} = \frac{1}{2 \times 10^3} = 0.5 \times 10^{-3} = 0.5mS$$

	V_1	V_2	V_3		
Node 1	$0.5m + 0.25m + 1m$	$-0.25m$	$-1m$	$=$	$20m$
Node 2	-0.25	$1m$	-0.25		$5m$
Node 3	$-1m$	-0.25	$1.25m$		$10m - 5m$



$$\text{KVL: } -60 + i_1(2) + i_1(1) - 1(i_2) = 0.$$

$$-1i_1 + 1i_2 + 3i_2 + 1i_2 - 1i_3 - 3i_4 = 0$$

$$\rightarrow -1i_2 + 1i_3 + 6i_3 + 4i_3 = 0$$

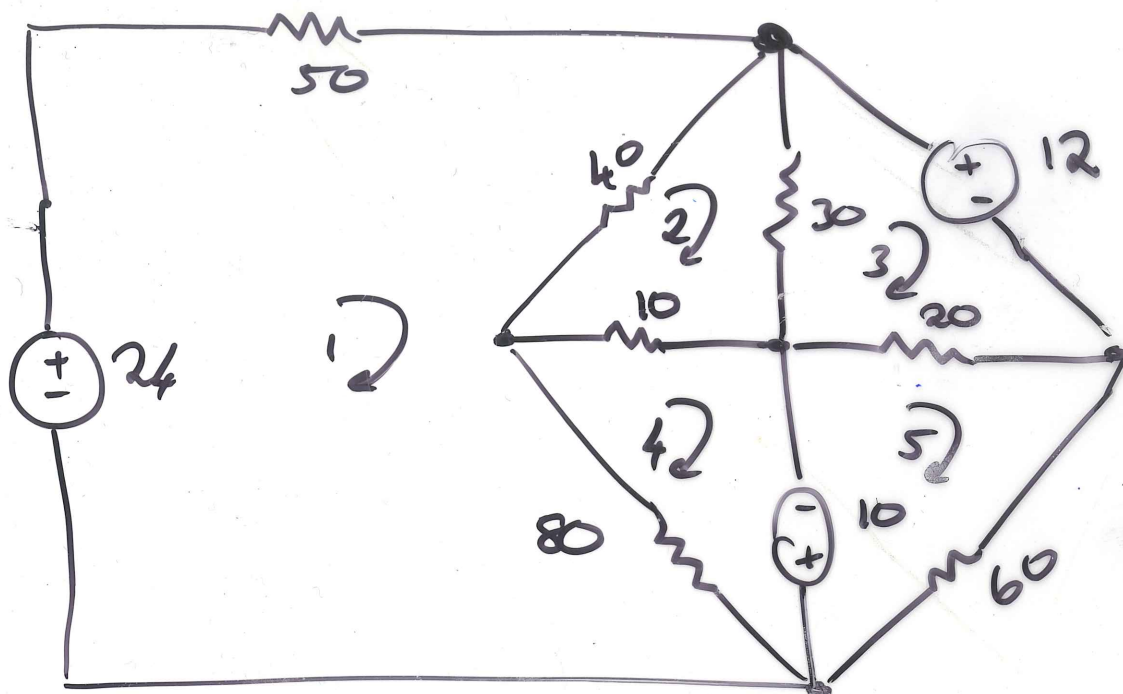
$$-10 + i_4(8) + 3i_4 - 3i_2 = 0$$

$$\rightarrow i_1(2+1) - i_2(1) + 0i_3 + 0i_4 = 60$$

$$-1i_1 + i_2(1+3+1) - i_3 - 3i_4 = 0$$

$$0i_1 - i_2 + i_3(1+6+4) + 0i_4 = 0$$

$$0i_1 - 3i_2 + 0i_3 + i_4(8+3) = 10$$



$$\begin{array}{c}
 \downarrow \\
 \begin{array}{c}
 i_1 \quad i_2 \quad i_3 \quad i_4 \quad i_5 \\
 \left[\begin{array}{ccccc}
 50+40+80 & -40 & 0 & -80 & 0 \\
 -40 & 40+30+10 & -20 & -10 & 0 \\
 0 & -30 & 30+20 & 0 & -20 \\
 -80 & -10 & 0 & 80+10 & 0 \\
 0 & 0 & -20 & 0 & 20+60
 \end{array} \right]
 \begin{array}{c}
 i_1 \\
 i_2 \\
 i_3 \\
 i_4 \\
 i_5
 \end{array}
 =
 \begin{array}{c}
 24 \\
 0 \\
 -12 \\
 10 \\
 -10
 \end{array}
 \end{array}
 \end{array}$$